
What is AWS Resource Access Manager (AWS RAM)?

AWS Resource Access Manager (RAM) is a service that allows you to **share AWS resources across accounts** and **within your AWS Organization** securely and efficiently.

It is especially useful in multi-account AWS environments where **centralized resources (like VPC subnets, Transit Gateways, License Manager configurations, etc.)** need to be **shared with other accounts** instead of duplicating resources in every account.

Why Use AWS RAM?

Without AWS RAM:

- Resources are **isolated per account** by default.
- Sharing usually means copying or duplicating resources.
- Centralizing infrastructure (like VPC, Route 53 Resolver rules) becomes hard.

With AWS RAM:

- **Share once, access many times.**
 - Reduces duplication.
 - Improves governance and cost efficiency.
 - Centralized resource management in multi-account architectures.
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Key Features of AWS RAM

Feature	Description
Cross-account Sharing	Share resources with specific AWS accounts or Organizational Units (OUs).
Organization-wide Sharing	Share with all accounts in an AWS Organization.
No Resource Duplication	Consumers use the same resource as the owner; no copies are made.
Tag-based Access Control	Manage sharing permissions via tags and IAM policies.
Support for Many Resource Types	Works with VPC subnets, Transit Gateways, Route 53 rules, etc.

Commonly Shared Resources

Here are the **resources** you can share using AWS RAM:

Resource Type	Description
VPC Subnets	Allow access to central VPCs (e.g., for shared services).
Transit Gateways	Enable inter-VPC or inter-account routing.
Route 53 Resolver Rules	Central DNS resolution across accounts.
License Manager Configs	Share licenses across accounts.
Resource Groups	Share resource collections across accounts.
AWS Outposts	Share local AWS infrastructure setups.

👉 The list is growing as AWS adds support for more services.

Core Concepts

Term	Meaning
Resource Owner	Account that owns the resource and initiates the sharing.
Resource Share	A container that defines the resource(s) being shared + recipients.
Principal	The recipient AWS account, OU, or Organization.
Accepted Share	The receiving account must accept the share (unless auto-accepted).

How It Works – Flow Diagram (Textual)

1. **Account A (Owner)** creates a Transit Gateway.
 2. In **AWS RAM**, Account A:
 - Creates a **Resource Share**.
 - Adds the Transit Gateway to it.
 - Specifies **Account B** as a principal (recipient).
 3. **Account B (Consumer)** accepts the share.
 4. Account B can now **attach VPCs to the shared Transit Gateway** from Account A.
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Use Cases

Scenario	Benefit
Centralized VPC architecture	One account manages networking; others consume shared VPCs.

Shared services (e.g., logging, DNS)	Central account shares resources like Route 53 resolver rules.
Shared Transit Gateway for interconnectivity	Central hub account manages routing.
Licensing control	Use AWS License Manager to enforce and share software licenses.
Hybrid cloud with AWS Outposts	Share Outposts across accounts.

Security & Access Control

- **IAM Policies** control who can create and manage shares.
 - Use **Tag-based policies** to enforce resource governance.
 - **Auto-accept sharing** is available when sharing within the same AWS Organization.
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Best Practices

1. Use **AWS Organizations** to simplify sharing across accounts.
 2. Enable **tagging and tag policies** for shared resources.
 3. Use **naming conventions** for Resource Shares (e.g., **Shared-VPC-Prod**).
 4. **Limit sharing to required principals** to reduce risk.
 5. **Monitor usage with AWS CloudTrail and Config**.
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Notes Summary

Topic	Details
Service	AWS Resource Access Manager (RAM)
Purpose	Share AWS resources across accounts/orgs securely
Supports	VPC Subnets, Transit Gateways, Route 53 rules, and more
Key Concepts	Resource Owner, Resource Share, Principals
Who Should Use It	Organizations with multiple AWS accounts
Security	IAM Policies, Tag-based control, Org-level auto-accept
Billing	Resources are billed to the owning account
